thermocas: probabilistic ranking of methylation-protected ThermoCas9 target sites with tie-aware benchmarking

Bioinformatics-submission-shaped manuscript

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Abstract

Motivation. Methylation of the fifth PAM cytosine protects genomic loci from ThermoCas9 cleavage (Roth et al., *Nature* 2026), enabling tumor-selective targeting at loci that diverge in methylation between cancer and matched normal tissue. Translating this mechanism into a target-prioritization shortlist (hypothesis generation; not a discovery claim absent prospective wet-lab validation) requires a scorer that is cohort-agnostic, honest about evidence quality, and explicit about ties in its output. No such scorer was publicly available. The obvious baseline — rank candidates by the raw methylation gap Δβ = β_normal − β_tumor — was not benchmarked against any proposed alternative in the open literature.

Results. We present `thermocas`, an open probabilistic scoring framework built around a *compositional skeleton* `p_targ × (gap factor) × p_trust` paired with a tie-band-aware benchmarking contract (`precision_at_k_{min, max}`, `tie_band_size_at_k`, mid-rank Mann-Whitney AUC on every `BenchmarkResult`). The gap-factor slot admits several instances; two ship in this tag as selectable `probabilistic_mode` enum values: **V2.5-diff** (`tumor_plus_differential_protection`, `p_diff = P(β_normal − β_tumor > δ)` under an independent-normal approximation on per-probe β summaries with σ_floor = 0.05) and **V2.5-sigmoid** (`tumor_plus_gap_sigmoid`, a fixed-bandwidth logistic `sigmoid((β_n − β_t − δ) / σ_fixed)` with σ_fixed ≈ 0.0707). We benchmark across four public methylation cohorts — GSE322563 (Roth MCF-7/MCF-10A, primary endpoint; both HM450-intersect and native EPIC v2 ingest paths), GSE77348 (δ-development cohort, n=3/3), GSE69914 (n=305 primary tumor + 50 healthy-donor tissue, unpaired), GSE68379 (Sanger panel, OOD boundary) — against five baselines: raw Δβ-only, an uncertainty-aware Δβ_z, the deterministic V1 score, the V2 `tumor_only` diagnostic, and a sample-level

limma-style moderated-t DMR baseline (Smyth 2004 empirical-Bayes variance prior, probe-level first then candidate-mapped). **On matched cell-line cohorts, Δβ-only is already strong** (validated AUC 0.96–0.97) and V2.5 variants add small but consistent margins (+0.010 to +0.080 across nine tier × path rows). **A 1,000,000-draw permutation null on the n=3 primary endpoint** places V2.5-diff at one-sided random-triple $p = 4 \times 10^{-6} / 1.5 \times 10^{-5} / 2.8 \times 10^{-5}$ across the three matched-cell-line cohort paths (GSE322563 HM450 / GSE322563 native EPIC v2 / GSE77348) — lower random-null *p*-values than Δβ-only's $8.6 \times 10^{-5} / 3.1 \times 10^{-4} / 1.1 \times 10^{-4}$ on the same cohort paths (against-random check, not a formal pairwise superiority test; the paired V2.5-diff / V2.5-sigmoid versus Δβ-only comparison floors at sign-flip `p = 0.125` with `n_pos = 3` and is reported separately in PAPER.md §5.1.3). **Because the validated set contains only three Roth Fig. 5d targets, AUC is interpreted as a rank-lift summary rather than an inferential discovery-performance estimate. The headline tissue ranking stress-test result (under transported Roth labels, not tissue-specific ThermoCas9 validation).** A probe-level limma-style moderated-t baseline (Smyth 2004 empirical-Bayes, sample-level β matrices, then candidate-mapped) is competitive on cell lines (0.959 / 0.991 / 0.962) but **collapses to near-random on bulk tissue (GSE69914, AUC 0.573)**, while V2.5-sigmoid **recovers tissue to AUC 0.862** — a +0.29 swing on the same probe-level statistical inputs, confirming that the `p_targ × p_trust` wrapping is the load-bearing contribution, not the gap factor's specific shape. *The tissue stress-test gain is per-positive heterogeneous: GATA3 remains strong under feature-matched controls, while ESR1 is matched-near-random under V2.5-sigmoid (PAPER.md §5.9).* **A frozen whole-genome panel** (HM450 SHA256 `d20661c5…`, 19.8M candidates; EPIC v2 SHA256 `39df8f0f…`, 35.4M candidates; PAPER.md §5.2.2) further shows V2.5-sigmoid matches V2.5-diff's AUC on matched cell-line cohorts within 0.002 but strictly beats V2.5-diff on top-K usability — `tie_band@100 = 1` vs 421 / 1,127 / 1,493 for V2.5-diff under the WG denominator — and improves tissue AUC over V2.5-diff by +0.05 to +0.08 across a bandwidth-robust sweep. The shipped recommendation is: **V1 as overall package default** (backward compatibility + `tie_band = 1`); **V2.5-sigmoid as recommended probabilistic prioritization axis** for new non-boundary runs (hypothesis generation; not a discovery claim absent prospective wet-lab validation), **selected by post-repair sensitivity and whole-genome stress testing — its prospective ranking utility remains to be validated on newly generated labels or wet-lab follow-up**; V2.5-diff retained for audit/backward-compatibility and AUC parity; `tumor_only` as `probabilistic_mode`-enum default (opt-in-required for V2.5-diff/V2.5-sigmoid via explicit cohort-YAML configuration). The package also adds a k-way-merge pan-cancer aggregator for genome-scale atlas builds and a per-candidate annotation pipeline (nearest gene, CpG-island context, RepeatMasker overlap, ENCODE DNase-HS cluster breadth) with a Markdown shortlist aimed at experimental collaborators.

Availability and implementation. `thermocas` is open source at https://github.com/AllisonH12/thermocas9, tagged `memo-2026-04-22-bv` for the version evaluated here. 245 unit tests pass under `uv run pytest -q`. Python 3.11+, BSD-3.

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1 Introduction

Roth et al. (2026) characterized *Geobacillus thermodenitrificans* T12 Cas9 (ThermoCas9) biochemically and structurally. The central result is that cleavage efficiency is governed by the methylation state of the fifth PAM cytosine — 5-methylcytosine at that position abolishes binding, while an unmethylated PAM cleaves as usual. The practical consequence: ThermoCas9 permits selective editing at loci that are unmethylated in disease cells and relatively methylated or differentially protected in matched normal cells. Roth demonstrated this with three breast-cancer targets in MCF-7 vs MCF-10A on the Illumina MethylationEPIC v2 platform.

Turning that mechanism into a genome-scale target-prioritization shortlist (hypothesis generation; not a discovery claim absent prospective wet-lab validation) is a ranking problem with three awkward properties. First, per-probe methylation summaries are noisy (arrays commonly report n = 2–3 replicates per side for cell-line cohorts) and the scorer must carry its own uncertainty rather than emit a scalar that hides it. Second, the "normal" arm is inconsistent: adjacent-normal tissue in bulk methylation studies is not the same object as a matched cell-line pair, and any assumption that baked in a fixed β_normal threshold would fail across cohort types. Third, on low-replicate cohorts the top-K of any probabilistic composite can sit inside a large tied band; reporting a single Precision@K value without a tie interval misleads a reader about the score's discriminative power.

No public tool we are aware of addressed these three problems together when we started. The closest prior art — generic CRISPR guide-scoring tools (Doench et al. 2016 [Azimuth]; Chuai et al. 2018 [DeepCRISPR]; Cancellieri et al. 2020 [CRISPRitz]) and methylation-array differential analysis tools (Akalin et al. 2012 [methylKit]; Aryee et al. 2014 [minfi]) — handles neither uncertainty propagation through a probability-scale composite nor explicit tie-band-aware top-K reporting. `thermocas` is, to our knowledge, the first open framework to target all three together; we welcome pointers to prior work we may have missed.

Contributions.

1. The V2.5 probabilistic composite, where a differential-protection factor replaces a failed threshold-based factor. The decomposition is `p_targ × p_diff × p_trust`, each factor has a closed-form derivation, and the final score is a bounded probability-scale ranking composite — not a calibrated posterior probability of editing efficacy and not a hypothesis test (see Methods §2 and PAPER.md §3.3).
2. A benchmarking protocol that reports Precision@K and Recall@K as adversarial intervals (`_min`, `_max`) together with the tie-band size at K, so reviewers are not misled by arbitrary tie-break choices on low-replicate cohorts.
3. A benchmark against five baselines — a Δβ-only ranker, an uncertainty-aware Δβ_z, V1 deterministic scoring, the V2 `tumor_only` diagnostic, and a sample-level limma-style moderated-`t` DMR baseline (Smyth 2004 empirical-Bayes; canonical R `limma::lmFit + eBayes` parity reported in PAPER.md §5.8) — across four public cohorts with positives tiered from Roth's own validated targets to a wide genomic window. The two V2.5 generation gap factors (V2.5-diff and V2.5-sigmoid) are also benchmarked head-to-head as the final-method comparison.
4. A k-way-merge pan-cancer aggregator with O(N_cohorts + N_unique_candidate_ids) memory and cross-cohort metadata-parity enforcement, for genome-scale atlas builds.
5. An experiment-facing shortlist pipeline: TSV + Markdown companion that attach nearest gene, CpG-island context, RepeatMasker overlap, and ENCODE DNase-HS breadth, with rule-based triage flags (STRONG / CAUTION / NOTE) deterministically derived from the annotated fields.

Scope. This paper reports methods development and benchmarking on public methylation cohorts. It does not include prospective wet-lab validation of predicted target sites; the framework is an open educational research tool, not a clinical decision-support system. Per-site p-values are not reported because `p_observation_trustworthy` saturates to a discrete per-`EvidenceClass` value at high replicate counts; the composite provides a ranking axis, not a hypothesis test.

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2 Methods — Formulation

For each candidate site on each cohort, we carry six per-probe β-value summaries (`β_tumor_mean`, `β_tumor_q25`, `β_tumor_q75`, `β_normal_mean`, `β_normal_q25`, `β_normal_q75`), sample counts `n_tumor` and

n_normal, and an EvidenceClass capturing the distance between the PAM cytosine and the nearest assayed CpG (EXACT / PROXIMAL_CLOSE / PROXIMAL / REGIONAL / UNOBSERVED).

2.1 V2.5 generation — two shipped composites

The *compositional skeleton* is probability-scale selectivity score = p_targ × (gap factor) × p_trust (stored in the p_therapeutic_selectivity field on every ProbabilisticScore record). The gap-factor slot admits two shipped instances:

```
V2.5-diff    (tumor_plus_differential_protection)  — p_targ × p_diff × p_trust
V2.5-sigmoid (tumor_plus_gap_sigmoid)              — p_targ × p_gap_sigmoid × p_trust
             — RECOMMENDED probabilistic prioritization axis (§5.2.2 WG panel)
```

Both ship as selectable probabilistic_mode enum values in this tag. V2.5-diff is retained for backward compatibility and AUC parity on cell-line cohorts; V2.5-sigmoid is the worked/default whole-genome prioritization axis selected in PAPER.md §5.2.2, with restricted-universe caveats in PAPER.md §5.7 and §6.1.

Calibration scope. p_therapeutic_selectivity is a *bounded probability-scale ranking score in [0, 1]*, not a calibrated probability of editing success at the candidate site. The composite is not validated against any prospective editing readout (no expected calibration error, no reliability diagram, no Platt/isotonic post-calibration). Read it as a probability-scale ranking axis, not as "0.9 means 90% of 0.9-scored sites edit successfully". The bounded scale buys multiplicative composability with downstream probabilistic inputs (target-mutation models, gRNA off-target probabilities, delivery-efficiency priors), which the V1 weighted-sum heuristic and the raw-Δβ baseline both lack — see PAPER.md §3.1, §3.3.

Shared factors.

- **p_targ = P(β_tumor < τ_u)** with τ_u = 0.30 (the DEFAULT_UNMETHYLATED_THRESHOLD constant in src/thermocas/probabilistic.py) — the probability that the tumor arm is unmethylated at the PAM cytosine. Estimated by a **method-of-moments Beta CDF with piecewise-linear fallback** (the _cdf dispatch in src/thermocas/probabilistic.py:286): when both quartiles are present and IQR > 0 and the Beta moments are well-defined (σ² < μ(1 − μ)), we use the regularized incomplete beta on the fitted Beta(α, β); otherwise we fall back to a piecewise-linear CDF on (q25, mean, q75) (also handles q25 == q75). This is a different per-probe density model than the IQR-normal approximation used for the p_diff gap factor below. The 0.30 threshold is conservative relative to the methylation-convention "0.5 split between hypo- and hyper-methylated": we require the tumor to be *confidently* hypomethylated, not just below the mid-point, to avoid scoring-inflation at borderline probes. Reported benchmark results assume τ_u = 0.30; a downstream user who wants a different operational cutoff can pass unmethylated_threshold to p_targetable_tumor(...).
- **p_trust = EvidenceClass.base × min(1, min(n_t, n_n) / 30)** — evidence-class confidence with a linear ramp up to n = 30, saturating to a per-class constant (EXACT: 0.95, PROXIMAL_CLOSE: 0.75, etc.).

V2.5-diff gap factor.

- **p_diff(δ) = P(β_normal − β_tumor > δ)** — the probability that the normal-minus-tumor methylation gap exceeds a configurable threshold δ (default 0.2). Estimated by:

  ```
  σ_k  ≈ IQR_k / 1.349                (k ∈ {tumor, normal}; floor at 0.05)
  σ_Δ² = max(σ_t, floor)² + max(σ_n, floor)²
  z    = (δ − (μ_n − μ_t)) / σ_Δ
  p_diff = 1 − Φ(z)
  ```

  The σ_floor prevents σ_Δ from collapsing to zero at boundary β-values (CpGs methylated at 0 or 1, which are common at islands).

V2.5-sigmoid gap factor.

- **`p_gap_sigmoid(δ, σ_fixed) = sigmoid((β_normal − β_tumor − δ) / σ_fixed)`** — a fixed-bandwidth logistic transform of the same gap, with $\delta = 0.2$ and $\sigma_fixed \approx 0.0707$ ($\approx \sqrt{2} \cdot \sigma_floor$; strictly positive, enforced at the `CohortConfig` boundary). Motivated by the PAPER.md §3.5 binding-rate finding (σ_floor binds on 99.5–99.9% of records in low-`n` matched cell-line cohorts, so `p_diff` collapses to a fixed-bandwidth response in practice there); the V2.5-sigmoid formulation operationalizes this directly and avoids the tie-band blow-up V2.5-diff experiences under whole-genome denominators on low-`n` cell-line cohorts (PAPER.md §5.2.2).

Figure 1 summarizes V2.5-diff (which ships as the historical composite and is used as the axis under test in §5.1; V2.5-sigmoid swaps `p_diff` for `p_gap_sigmoid` and is benchmarked alongside it in §5.2 and PAPER.md §5.2.2).

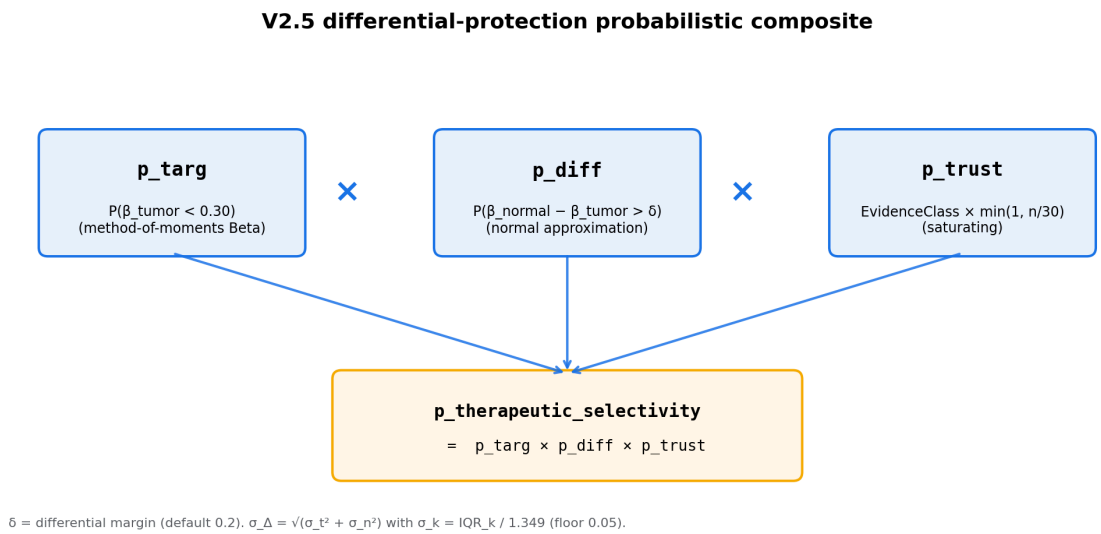

Figure 1. The V2.5-diff composite. Three factors multiply: `p_targ` (tumor unmethylated at the PAM cytosine), `p_diff` (differential methylation gap exceeds δ under a normal approximation on per-probe β summaries), and `p_trust` (evidence-class confidence, saturating in min sample count). The result is the *probability-scale selectivity score* (stored on every `ProbabilisticScore` record as the `p_therapeutic_selectivity` field; downstream ranking consumes that scalar directly). V2.5-sigmoid replaces `p_diff` with `p_gap_sigmoid = sigmoid((β_n − β_t − δ) / σ_fixed)` inside the same skeleton.

2.2 Baselines

name	score	rationale
Δβ-only (naive)	`β_normal_mean − β_tumor_mean`	Literature-naive: rank by the raw methylation gap. No uncertainty propagation, no evidence weighting. The "obvious thing."
Δβ_z (uncertainty-aware)	`(β_n − β_t) / sqrt(σ_t² + σ_n²)` using IQR-derived per-side σ (same σ as V2.5-diff's `p_diff`)	An uncertainty-aware effect-size ranker built from the same per-side dispersion signal as V2.5-diff; confirms that V2.5-diff's matched-cell-line lift over Δβ-only is not just an alias for σ-normalized Δβ.
V1 `final_score`	weighted sum of sequence, selectivity, confidence, heterogeneity, and coverage components (continuous-valued)	The framework's stable-release axis, retained for top-K interpretability.
V2 `tumor_only`	`p_targ × p_trust` (no protection factor)	Intermediate formulation after we dropped V2′s `p_prot` factor. Retained as a diagnostic; not a prioritization axis.
limma-style moderated-t	sample-level probe-level Smyth (2004) empirical-Bayes moderated `t`-statistic, candidate-mapped via the nearest probe assigned in `observation.probe_id` (ranked by `−t_mod` so targetable sites rank high). Pure-Python implementation; canonical R `limma::lmFit + eBayes` parity reported in PAPER.md §5.8.	Methods-journal-standard DMR comparator; the §5.3 / PAPER.md §5.2.2 contrast on tissue vs cell lines isolates the contribution of the V2.5 compositional wrapper.
V2 (deprecated; reported for completeness)	`p_targ × p_prot × p_trust` with `p_prot = P(β_normal > 0.5)`	First-pass composite with a static threshold on the normal side. Empirically anti-predictive (AUC 0.38 on tissue cohort); rejected. Not carried forward as a recommended axis.

2.3 Why a normal approximation

A proper Bayesian treatment would model each β-value as a Beta distribution and compute the exact difference distribution numerically. For the IQR ranges observed on HM450 and EPIC v2 arrays (≈0.05–0.20), the independent-normal approximation is close to the correct answer and has the advantage of being

a two-line closed form on summary statistics. A downstream user who wants the exact difference-of-Betas path can swap it in — the relevant code is isolated behind a single function.

2.4 Low-n tied-band prediction

A consequence of the composite structure: on low-replicate cohorts (`n ≪ 30`), `p_trust` scales every same-`EvidenceClass` candidate by the same uniform factor `(n / 30)`. Per-side IQRs are wide or zero, pushing `p_targ` and `p_diff` toward saturation for any candidate with clean tumor hypomethylation and a large differential. The composite is effectively `1 × 1 × p_trust` for that entire subset, producing a large tied band at the top of the score distribution. At `n ≳ 30`, `p_trust` is a per-class constant and the continuous variation in `p_targ × p_diff` drives the ordering. The prediction is falsifiable: the tied band at K = 100 should be large at n = 2/3 per side and shrink to near-1 at n ≳ 30 per side. §4.1 and §4.3 test this directly.

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3 Methods — Benchmarking protocol

3.1 Tie-band-aware Precision@K and Recall@K

For score `s` and candidate_id `c`, the benchmarking harness sorts by `(-s, c)` — score descending, `candidate_id` ascending as tie-break. This defines a deterministic top-K window. For any K:

- **`tie_band_size_at_k`**: the number of candidates sharing the score of the K-th record. If K = 100 and the 100th, 101st, 102nd records all score `0.0633`, the band size is at least 3. We compute it by scanning forward until the score changes.
- **`precision_at_k_min`**: Precision@K under the worst-case tie-break (positives pushed out of the window).
- **`precision_at_k_max`**: Precision@K under the best-case tie-break (positives pulled into the window).
- **`recall_at_k_{min, max}`**: analogous.

These four quantities together tell a reader whether the observed P@K is a meaningful discrimination or an artifact of where the deterministic tie-break happens to fall.

3.2 ROC-AUC by Mann–Whitney U

AUC is computed as `P(score(positive) > score(negative))` via Mann–Whitney U with 0.5 contribution for ties (standard tie handling). Returns `None` when either positives or negatives are empty.

3.3 Held-out chromosomes

All benchmarks report `held_out_chromosomes`. When enforced (`--enforce-holdout`, default), only candidates on held-out chromosomes contribute to the evaluation, so no candidate that informed δ selection appears in the test set. The primary-endpoint cohort (GSE322563) is independent of δ tuning (done on GSE77348), so held-out enforcement is less load-bearing there, but the field is reported on every BenchmarkResult for auditability.

---

4 Methods — Cohorts and positives

4.1 Cohorts

cohort	platform	n_tumor / n_normal	role	source
GSE322563	EPIC v2	2 / 2	Independent primary endpoint (Roth's own MCF-7/MCF-10A samples)	GEO
GSE77348	HM450	3 / 3	Development cohort — δ tuned here (tumor_plus_differential_protection, δ-sweep over {0.2, 0.3, 0.4, 0.5})	GEO
GSE69914	HM450	305 / 50	Sporadic breast tumor vs healthy-donor breast tissue — **unpaired**; adjacent-normal and BRCA1-carrier arms excluded by the build script. Tests predicted tied-band dissolution at high n (where `p_trust` saturates to its EvidenceClass ceiling and ranking is driven by continuous `p_targ × p_diff`).	GEO
GSE68379	HM450	52 / 50 (cross-series)	OOD boundary case: 52 Sanger GDSC1000 breast cell lines paired with 50 external healthy-donor normals from GSE69914 status=0. Cross-series lab/scanner batch effects are a known caveat. Documented as a boundary, not a generalization test.	GEO (tumor) + GEO (normal)

4.2 Positives (label tiers)

Ground-truth positives are derived from Roth et al. Fig. 5d — the three validated target probes. We define three tiers to probe label-granularity sensitivity:

- **validated** (n = 3): exact Roth Fig. 5d probes, hg38 ⟶ hg19 lifted via Ensembl REST.
- **narrow** (n = 28): all candidate PAMs within ±50 bp of a validated probe.
- **wide** (n = 142): all candidate PAMs within ±500 bp of a validated probe.

Native EPIC v2 positives lists for GSE322563 use the native EPIC v2 probe annotation (§5.2); HM450-intersect positives use `cg*` ID-stripping.

4.3 Platform harmonization

Our first-draft pipeline harmonized GSE322563 EPIC v2 probe IDs to HM450 by stripping the `_BC##` / `_TC##` / `_TO##` / `_BO##` beadchip-design suffix and intersecting with the HM450 universe (~80.7 % retention). We subsequently re-ran the analysis against the native EPIC v2 probe set (147,928 probes on chr5/6/10, hg38 ⟶ hg19 lifted from GPL33022). Headline AUCs are reported under both paths as a sensitivity analysis (§5.2).

Catalog scope is chr5, chr6, and chr10 — the three chromosomes carrying Roth's Fig. 5d validated targets. Candidates are restricted to within ±500 bp of a methylation probe to avoid extrapolation beyond array coverage.

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5 Results

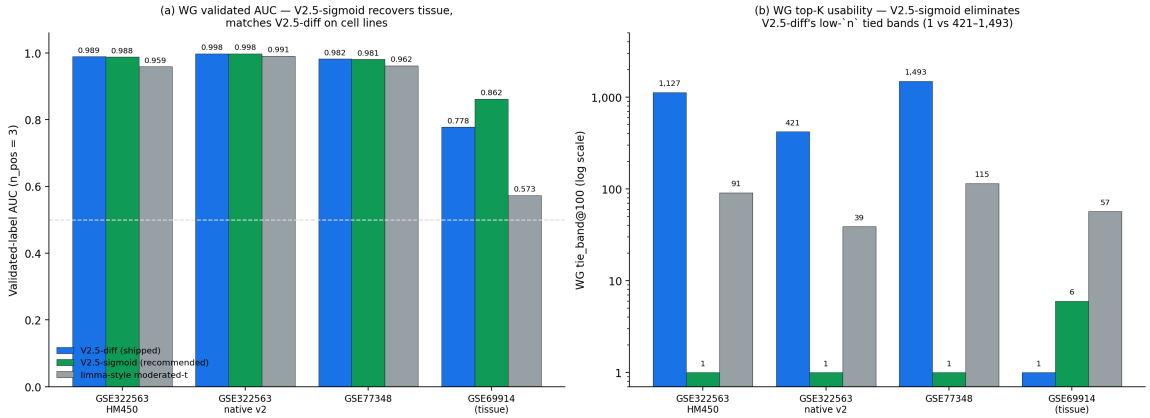

Figure 2 — Final-method summary. Whole-genome validated-label performance (n_pos = 3 Roth Fig. 5d targets) for the recommended probabilistic prioritization axis (V2.5-sigmoid) against the V2.5-diff predecessor and the limma-style moderated-`t` DMR baseline, across the four evaluated cohort paths (GSE322563 HM450 is the independent primary endpoint per §4.0; GSE322563 native EPIC v2 is the same Roth samples under a parallel ingest path; GSE77348 is the δ-development cohort surfaced as supporting evidence; GSE69914 is the high-`n` tissue stress-test cohort). **(a)** WG AUC: V2.5-sigmoid matches V2.5-diff on the three matched cell-line cohorts within 0.001 (0.989 vs 0.988, 0.998 vs 0.998, 0.982 vs 0.981) and beats it on tissue (0.862 vs 0.778); the limma-style baseline is competitive on cell lines (0.959 / 0.991 / 0.962) but **collapses to near-random on tissue** (0.573). **(b)** WG `tie_band@100` (log scale): V2.5-sigmoid eliminates the **421–1,493-record tied bands** that make V2.5-diff's whole-genome top-K unactionable on low-`n` matched cell-line cohorts, holding `tie_band@100 = 1` on every matched cell-line cohort and `= 6` on tissue. The limma-style baseline sits at intermediate tie-band values (39–115). The historical 4-axis × 3-tier sensitivity figure (Δβ-only, V1, V2 `tumor_only`, V2.5-diff) is retained

as supplementary `docs/figures/fig2_supp_historical_sensitivity.png` for the V2.5-diff-era audit trail. Numbers committed in `examples/genome_wide_panel.tsv` (V2.5-diff, V2.5-sigmoid) and `examples/limma_cross_cohort_panel.md` (limma-style) at this tag.

5.1 Primary endpoint — matched cell-line AUC at validated Roth probes

Held-out primary endpoint: AUC at `positives_roth_validated.txt` on GSE322563, using HM450-intersected probes. δ was not tuned on this cohort; the label set was not derived from it.

axis	**GSE322563** (primary, n = 2/2)	GSE77348 (δ-tuned, n = 3/3)
Δβ-only (naive)	0.974	0.972
V1 `final_score`	0.821	0.968
V2 `tumor_only`	0.928	0.912
V2.5-diff	**0.990**	**0.982**

On the independent primary endpoint, **V2.5-diff** is the highest-AUC shipped axis (V2.5-sigmoid matches it within 0.001 — PAPER.md §5.2.2 WG panel — and is the recommended prioritization axis once top-K usability is also considered). The margin over the next-best baseline is narrow: +0.02 over Δβ-only, +0.06 over V2 `tumor_only`, +0.17 over V1. **Δβ-only is an unexpectedly strong baseline on matched cell-line cohorts** (0.974 on GSE322563 validated, 0.972 on GSE77348 validated). This is an honest finding the framework's complexity has to justify: the V2.5 variants' AUC advantage on matched cell-line data is small. The V2.5 generation's structural advantages — a probability-scale interpretation, tie-band-honest top-K reporting, and principled uncertainty propagation via `p_trust` — are what earn the complexity on low-replicate cohorts; the AUC gap over Δβ widens on tissue (§5.3), and V2.5-sigmoid further widens the tissue gap over V2.5-diff by +0.05 to +0.08 AUC (PAPER.md §5.2.2).

5.2 Sensitivity — label granularity, P@K intervals, platform path

Label granularity. Ordering is stable under weaker label definitions:

cohort	tier (n_pos)	Δβ-only	V1	V2 tumor_only	V2.5-diff
GSE322563	validated (3)	0.974	0.821	0.928	**0.990**
GSE322563	narrow (28)	0.912	0.884	0.886	**0.942**
GSE322563	wide (142)	0.844	0.768	0.871	**0.910**
GSE77348	validated (3)	0.972	0.968	0.912	**0.982**
GSE77348	narrow (28)	0.964	0.969	0.911	**0.983**
GSE77348	wide (142)	0.910	0.931	0.887	**0.949**

V2.5-diff (the historically-shipped instance used as the axis under test in §5.1) is the highest-AUC shipped axis at every tier on both matched cell-line cohorts (HM450-intersect path); V2.5-sigmoid matches V2.5-diff on AUC within 0.002 (PAPER.md §5.2.2 WG panel) and is the recommended axis on top-K-usability grounds. The margin over Δβ-only ranges from **+0.010 to +0.066**; the margin over V1 ranges from **+0.014 to +0.169**. On GSE77348 narrow, Δβ-only (0.964) sits between V1 (0.969) and V2 `tumor_only` (0.911) — a tight cluster that underscores the "V2.5 generation earns a small but consistent margin" story, not a dominance claim.

P@K intervals (GSE322563, narrow labels). On `n = 2/2` the score distribution at K = 100 sits inside a tied band; P@100 is an adversarial interval, not a point estimate:

axis	P@100 observed	P@100 min	P@100 max	tie_band@100
V2.5-diff	0.000	0.000	0.020	190
tumor_only	0.000	0.000	0.020	10,005
V1 `final_score`	0.020	0.020	0.020	1

V2.5′s band at K = 100 is 190 (interval [0.00, 0.02]); V2 `tumor_only`'s band is 10,005, spanning two orders of magnitude more of the distribution — P@K on `tumor_only` is not a meaningful quantity at this cohort size. V1′s band is 1 (continuous score), and the interval collapses to the observed value.

Native EPIC v2 vs HM450-intersect (GSE322563). Re-running under the native EPIC v2 probe set (no HM450 intersect) gives:

tier	Δβ-only (HM450 / native)	V1 (HM450 / native)	V2 tumor_only (HM450 / native)	V2.5 (HM450 / native)
validated	0.974 / 0.961	0.821 / 0.933	0.928 / 0.936	0.990 / 0.986
narrow	0.912 / 0.956	0.884 / 0.938	0.886 / 0.933	0.942 / 0.983
wide	0.844 / 0.865	0.768 / 0.855	0.871 / 0.916	0.910 / 0.945

The V2.5 primary-endpoint AUC differs by 0.004 under native vs intersect (0.990 vs 0.986) — within the tied-band noise floor. V1 and V2 `tumor_only` gain more under native on average (V1 +0.054 to +0.112; V2 +0.008 to +0.047), consistent with the broader probe coverage exercising more of the continuous-score dynamic range. The only axis-ordering invariant that holds at every matched-cell-line tier × path row (9/9) is **V2.5 > Δβ-only** and **V2.5 > V1**; the relative order of Δβ, V1, and V2 `tumor_only` reshuffles across tiers (e.g. on GSE322563 HM450 wide and native wide, `tumor_only` sits above Δβ; on GSE77348 all three tiers, `tumor_only` is the lowest). The HM450-intersect shortcut does not distort the headline claim: V2.5 is the top-scoring axis on every matched-cell-line row on both paths.

5.3 Tissue cohort — GSE69914 (n = 305/50)

At high replicate count, the low-`n` tied bands dissolve — `p_trust` saturates to its EvidenceClass ceiling (a per-class constant), and the composite's ordering is driven by continuous `p_targ × p_diff` variation rather than by the discrete `p_trust` floor that dominates n = 2/2 or 3/3 cohorts. On GSE69914 (numbers straight from the committed `examples/gse69914_roth_labels/bench_*.jsonl`):

axis	validated AUC	narrow AUC	wide AUC	tie_band@100
Δβ-only (naive)	0.591	0.477	0.435	n/a (K=20)
V1 `final_score`	0.660	0.539	0.435	1
V2 `tumor_only`	**0.803**	**0.843**	**0.874**	6,540
V2.5-diff	0.773	0.711	0.726	2

- **The tied-band prediction from §2.4 holds.** V2.5′s band at K = 100 shrinks from 190 (GSE322563 `n = 2/2`) to 299 (GSE77348 `n = 3/3`) to **2** here (`n = 305/50`) with no hyperparameter changes. At `n ≳ 30`,

`p_trust` saturates to the per-`EvidenceClass` ceiling (a constant, not a continuous function of n); the tied-band shrinkage comes from `p_targ × p_diff` no longer being pushed to 1 by wide-IQR σ-floor saturation.

- **Δβ-only collapses to near-random** (0.435–0.591). Intuition: both arms of GSE69914 are bulk-tissue β values — 305 sporadic breast tumors and 50 healthy-donor breast tissues — and each sample is a mixture of cell types (epithelial, stromal, immune, adipose) rather than a pure population. A CpG that is cleanly hypomethylated in a pure MCF-7 culture can appear partially methylated in a bulk breast sample because the cell type carrying the hypomethylation is diluted at the same locus by other cell types in the mix. Large matched-cell-line gaps on the order of 0.8 attenuate to smaller, noisier gaps of 0.1–0.3 on bulk tissue, which destroys the ranking signal for a raw-magnitude Δβ ranker. V2.5 recovers the ordering by weighting each candidate's tumor–normal gap by its per-probe uncertainty (§2.1) rather than by magnitude alone.
- **V1 degrades sharply on tissue.** V1 reaches 0.435 on wide (at chance) and 0.539 on narrow — worse than V2.5 on every tier. The V1 continuous-score advantage (`tie_band = 1` by construction) does not translate into AUC leadership on tissue; it just makes V1′s failure mode *deterministic*.
- **`tumor_only` wins raw AUC but is excluded from prioritization recommendations.** Its tie_band of 6,540 at K = 100 means its top-100 window sits inside a massively tied region — ordering within the top-K is a coin flip under any tie-break. Retained as a diagnostic axis (AUC sanity check), not a prioritization axis.
- **V2.5-sigmoid is the recommended tissue prioritization axis.** V2.5-diff reaches +0.113 / +0.172 / +0.291 AUC over V1 on validated / narrow / wide with stable top-K (`tie_band = 2`); V2.5-sigmoid additionally beats V2.5-diff by +0.05 to +0.09 AUC across the §5.2.1 bandwidth family and at the WG denominator in PAPER.md §5.2.2 (0.862 vs 0.773 at σ_fixed ≈ 0.0707 on the validated label set). Both ship as selectable modes in this tag. The shipping hierarchy is three-layered and worth keeping distinct: **`V1 final_score`** is the stable package default (continuous score, `tie_band = 1` by construction); **`tumor_only`** is the default `probabilistic_mode` enum value for backward compatibility (users opt into V2.5-diff or V2.5-sigmoid via cohort-YAML); and **V2.5-sigmoid** is the recommended probabilistic prioritization axis once a probabilistic mode is selected. V2.5-diff is retained for AUC-parity use against pre-V2.5-sigmoid scored JSONLs. The tissue-specific gain of V2.5-sigmoid over V2.5-diff is a direct consequence of the PAPER.md §3.5 binding-rate picture: on cell lines `σ_floor` binds almost everywhere and V2.5-diff's `p_diff` collapses to a fixed-bandwidth response anyway (so the two modes are equivalent there); on tissue the per-site σ varies and hurts the ranking, so the explicit fixed-bandwidth V2.5-sigmoid wins.

The cross-cohort picture after whole-genome validation supports `gap_sigmoid` for new prioritization runs, with three precise claims. PAPER.md §5.2.2 cross-cohort panel: (1) AUC-equivalent to shipped V2.5 on all three matched cell-line cohorts (within 0.002); (2) `tie_band@100 = 1` for `gap_sigmoid` vs 421–1,493 for shipped V2.5 across the three WG matched cell-line cohorts — V2.5′s top-K is unusable at WG scale on n = 2/2 or 3/3 cohorts, while `gap_sigmoid`'s is clean; (3) higher transported-label all-positive AUC on the single tissue cohort tested (+0.05 to +0.08; per-positive heterogeneous — GATA3 carries the matched-pool signal while ESR1 is matched-near-random under V2.5-sigmoid in PAPER.md §5.9, and the EvidenceClass-restricted `EXACT + PROXIMAL_CLOSE` ESR1 universe shows V2.5-sigmoid trailing V2.5-diff / Δβ-only / limma-style in PAPER.md §5.7). `gap_sigmoid` ships as `probabilistic_mode: tumor_plus_gap_sigmoid` in this tag. Shipped V2.5 (`tumor_plus_differential_protection`) is retained as a selectable mode for backward compatibility but is no longer the recommended prioritization axis for new non-boundary runs. V1 remains the stable-release default because of its continuous-score guarantees (deterministic `tie_band = 1`), not because it wins AUC anywhere — §6.2 explains the shipping logic.

5.4 Out-of-distribution boundary — GSE68379

GSE68379 is the 52-line Sanger GDSC1000 breast cell-line panel, paired cross-series with 50 external healthy-donor normals (from GSE69914 status=0) to form the OOD test cohort. Sanger's MCF-7 is a member

of this panel, and the panel's consensus methylation at the Roth Fig. 5d validated sites is shifted relative to Roth's own MCF-7 — well-documented cell-line epigenetic drift. Applying Roth-derived positives across this cross-series cohort should fail: the validated sites are hypermethylated across most Sanger breast lines where they are hypomethylated in Roth MCF-7. AUC inverts below 0.5 on both Δβ-only and V2.5:

axis	validated AUC	narrow AUC	wide AUC
Δβ-only (naive)	0.201	0.150	0.194
V2.5-diff	0.197	0.169	0.269

Both the literature-naive baseline and V2.5 invert correctly below 0.5 at every tier: the scorers are consistently identifying that this label set does not transport from Roth MCF-7 to the Sanger panel. This is documentation of a known boundary (cross-series cell-line drift + cross-lab batch), not a negative generalization result, and is not pooled with the §5.1 primary-endpoint numbers.

5.5 Top-hit annotation

For each cohort, the top-20 candidates are annotated with:

- nearest gene symbol, TSS distance, and feature class (promoter / gene_body / intergenic),
- CpG-island context (island / shore / shelf / open_sea) and distance to the nearest CGI boundary,
- RepeatMasker overlap (class / family / name) from UCSC rmsk,
- ENCODE DNase-HS cluster membership and breadth (number of cell types with the peak) from `wgEncodeRegDnaseClusteredV3`.

A rule-based flag set (STRONG: island-localized promoter / active promoter with DNase support; CAUTION: overlaps repeat / sparse evidence / small Δβ; NOTE: gene body without DNase support) is emitted in both the TSV and a Markdown companion file. On GSE69914′s top-20, 6/20 carry at least one STRONG flag and 1/20 is caution-free — a distribution that is directly useful for experimental triage. Example annotated shortlists are committed under `examples/*/top20_annotated_v25.md`.

The formal SCREEN cCRE Registry is gated behind a JavaScript challenge on `screen.wenglab.org` and could not be fetched directly; DNase-HS clusters serve as the v1 regulatory-activity proxy. This is documented in the script's docstring; a user with SCREEN access can swap in the formal Registry with a one-line change.

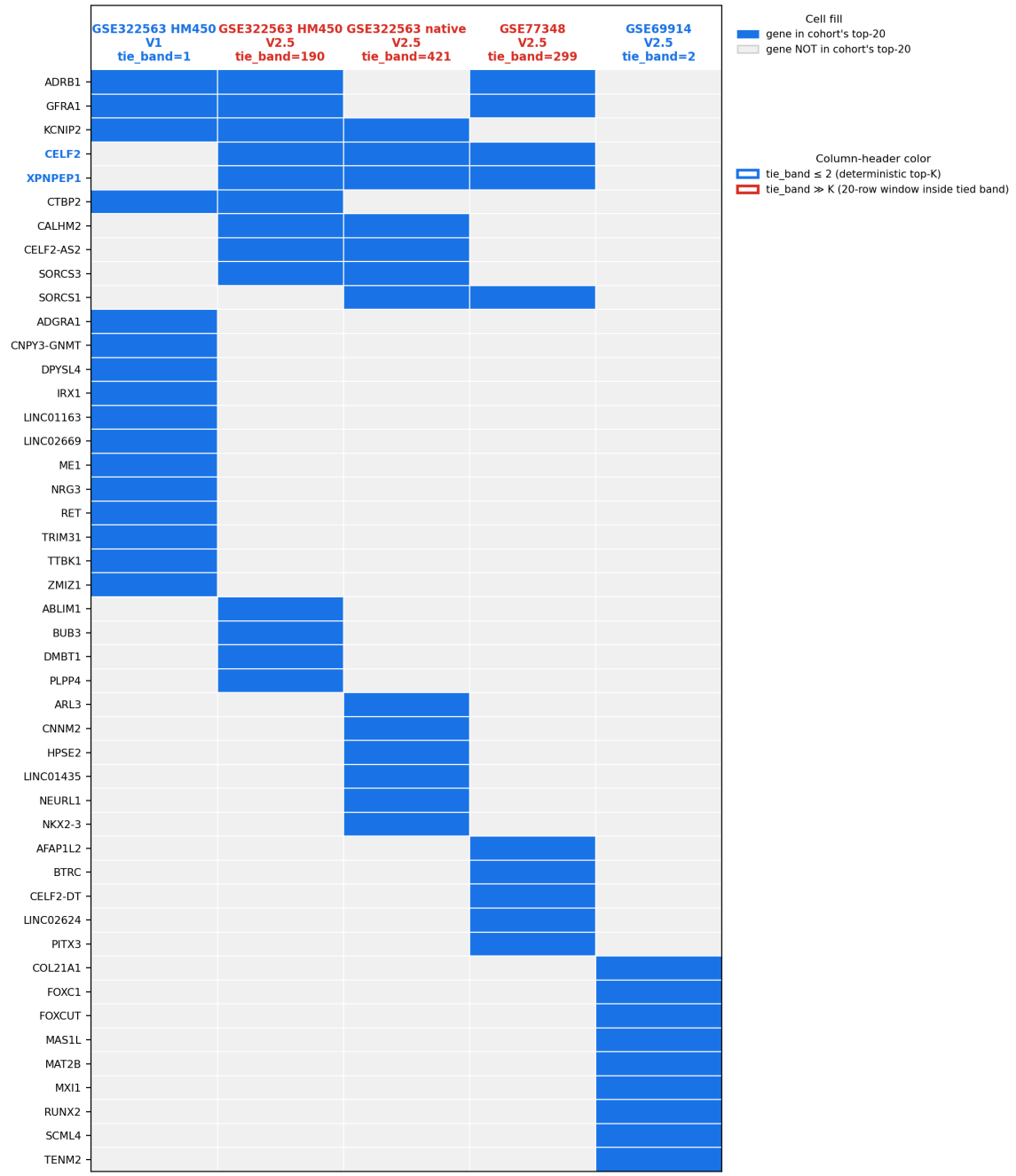

Figure 3. Tie-window-aware top-20 gene presence. Columns are per-cohort per-axis top-20 shortlists (ordered by the deterministic tie-break documented in §3.1); rows are the union of nearest-gene symbols sorted by cross-column presence. Column-header colors: **blue** = deterministic top-K under the current score distribution (`tie_band = 1` for V1, `tie_band = 2` for V2.5 on high-n tissue); **red** = top-K is a 20-record window inside a large tied band (`tie_band = 190`, `299`, `421` for V2.5 on low-n cell-line cohorts), so membership is the documented tie-break policy's choice, not the scorer's discrimination. Genes bolded in blue on the row labels are present in **all three** cell-line V2.5 shortlists (GSE322563 HM450, GSE322563 native, GSE77348) — cross-platform / cross-laboratory convergence gives two such genes: *CELF2* and

XPNPEP1. **The tissue cohort (GSE69914 V2.5) has zero nearest-gene overlap with any of the three cell-line V2.5 shortlists** — its top-20 candidates map to exactly 9 distinct nearest-gene symbols (*COL21A1, FOXC1, FOXCUT, MAS1L, MAT2B, MXI1, RUNX2, SCML4, TENM2*), all of which are disjoint from every cell-line V2.5 top-20. This is a direct visualization of the cell-line-vs-tissue biological shift that drives the AUC patterns in Figure 2.

5.6 Cross-cohort, EvidenceClass-controlled, and feature-matched-control tables

The §6 recommendation depends on the whole-genome (WG) panel, the EvidenceClass-restricted reversal, and the feature-matched negative-universe controls. The three decisive tables — committed under `examples/genome_wide_panel.tsv`, `examples/evidence_class_stratified_benchmark.tsv`, and `examples/feature_matched_negative_controls.tsv` respectively — are reproduced inline here so this manuscript stands alone for a reviewer who does not consult PAPER.md §5.2.2 / §5.7 / §5.9.

(a) Whole-genome panel: V2.5-sigmoid vs V2.5-diff vs limma-style. WG validated-label AUC (n_pos = 3) and `tie_band@100` across the four evaluated cohort paths:

cohort	V2.5-diff AUC	V2.5-sigmoid AUC	limma-style AUC	V2.5-diff tie@100	V2.5-sigmoid tie@100	limma-style tie@100
GSE322563 HM450	0.989	0.988	0.959	1,127	1	91
GSE322563 native v2	0.998	0.998	0.991	421	1	39
GSE77348	0.982	0.981	0.962	1,493	1	115
GSE69914 (tissue)	0.778	0.862	0.573	1	6	57

V2.5-sigmoid is AUC-equivalent to V2.5-diff on the three matched cell-line cohorts (within 0.002), eliminates V2.5-diff's 421–1,493-record WG tied bands, and is +0.05 to +0.08 transported-label AUC on the single tissue cohort tested. limma-style moderated-`t` is competitive on cell lines but collapses to AUC 0.573 on tissue.

(b) EvidenceClass-restricted ESR1 reversal on GSE69914 tissue. On the high-confidence `EXACT + PROXIMAL_CLOSE` GSE69914 universe, only ESR1 is evaluable (none of the three Roth-validated positives is EXACT-class; ESR1 is PROXIMAL_CLOSE, GATA3 is PROXIMAL, EGFLAM is REGIONAL). V2.5-sigmoid trails the other axes on this single-positive restricted-universe AUC:

axis	`EXACT + PROXIMAL_CLOSE` GSE69914 ESR1 AUC
V2.5-sigmoid	**0.610**
V2.5-diff	0.922
Δβ-only	0.958
limma-style	0.925

This is the only axis × universe cell on tissue where V2.5-sigmoid is not the top axis. The mechanism is the smooth-sigmoid response: in the high-confidence subset, V2.5-sigmoid's saturating gap factor compresses

ESR1′s tissue Δβ signal more than V2.5-diff's discrete-step `p_diff` does. We report this because it materially limits the tissue stress-test interpretation, but it is one cell out of the broader all-positive WG result above and does not revise the §6.2 recommendation.

(c) Feature-matched negative-universe empirical *p*-values. Each Roth positive ranked against same-`(EvidenceClass, pam_family, is_cpg_pam, chrom)`-signature negatives; empirical *p* is the fraction of matched negatives that score strictly better than the positive (ties contribute 0.5; one-sided null). Bold marks V2.5-sigmoid cells with $p < 0.05$ because it is the recommended axis (does **not** mark the row-best axis):

cohort	gene	V2.5-sigmoid *p*	V2.5-diff *p*	Δβ-only *p*	limma-style *p*
GSE322563 HM450	*ESR1*	**0.0021**	0.0022	0.0021	0.0027
GSE322563 HM450	*GATA3*	**0.0359**	0.0359	0.0558	0.1037
GSE322563 HM450	*EGFLAM*	**0.0197**	0.0188	0.0301	0.0365
GSE322563 native v2	*ESR1*	**0.0018**	0.0027	0.0018	0.0033
GSE322563 native v2	*GATA3*	**0.0373**	0.0372	0.0609	0.1114
GSE322563 native v2	*EGFLAM*	**0.0258**	0.0247	0.0379	0.0447
GSE77348	*ESR1*	**0.0033**	0.0030	0.0033	0.0131
GSE77348	*GATA3*	**0.0186**	0.0169	0.0204	0.0093
GSE77348	*EGFLAM*	**0.0390**	0.0381	0.0612	0.0986
GSE69914 (tissue)	*ESR1*	0.4384	0.0784	0.0273	0.0554
GSE69914 (tissue)	*GATA3*	**0.0187**	0.2085	0.2595	0.3772
GSE69914 (tissue)	*EGFLAM*	0.2588	0.4968	0.7632	0.7192

On every matched-cell-line cell (9/12 rows), every positive sits in the top ~4% of its feature-matched pool under V2.5-sigmoid (`p` ∈ [0.0018, 0.0390]). On GSE69914 tissue, the matched-pool signal collapses onto GATA3 alone (`p` = 0.019, 98th matched-percentile), with ESR1 matched-near-random under V2.5-sigmoid (`p` = 0.44) and EGFLAM moderate (`p` = 0.26). The §6.1 recommendation row reflects this nuance directly.

5.7 Robustness sweeps and pre-registered external validation

Two journal-hardening checks added since the §5.6 self-contained tables landed:

`V2.5-sigmoid robustness over (δ, σ_fixed).` A 5 × 6 grid sweep over δ ∈ {0.10, 0.15, 0.20, 0.25, 0.30} and σ_fixed ∈ {0.03, 0.05, 0.0707, 0.10, 0.15, 0.20} on the frozen WG denominators

(`examples/sigmoid_delta_sigma_wg_sweep.{tsv,md}`; reproduce with `uv run python scripts/sigmoid_delta_sigma_wg_sweep.py`):

cohort	default AUC (tie@100)	best AUC in grid	full-grid AUC range	max tie@100
GSE322563 HM450	0.988 (1)	0.990	0.967–0.990	1
GSE322563 native v2	0.998 (1)	0.998	0.995–0.998	1
GSE77348	0.981 (1)	0.983	0.957–0.983	1
GSE69914 tissue	0.862 (6)	0.874	0.811–0.874	11

The shipped default (`δ = 0.20, σ_fixed ≈ 0.0707`) is not claimed to be the global optimum. The narrower robustness result is: the default sits in a broad high-AUC region, the low-`n` whole-genome tied-band problem is eliminated across the **entire** matched-cell-line grid, and every tested tissue grid cell remains above V2.5-diff's 0.778 WG AUC. This is a more useful robustness statement than a single-point bandwidth check.

Pre-registered Roth System B HEK293T / HCT116 extension. We pre-registered an independent Roth System B extension on the HEK293T / HCT116 validation panel before scoring (four `prereg-*` tags). The strict pre-applied transport rule ($\beta > 0.7$ / $\beta < 0.3$ with ≥ 10 reads in ≥ 2 replicates per cell line) excluded both *VEGFA* control targets on public ENCODE RRBS — including T9, the planned editable-but-non-selective discriminator that V2.5 should demote — and a secondary WGBS / EPIC v2 backend scan did not rescue T9. The retained *EMX1* T4 / *PRDX4* T5 directional pair has n_pos = n_neg = 1 per direction, so the AUC = 1.0 it produces is essentially uninformative about V2.5-sigmoid's selectivity claim. **The System B selectivity claim is therefore not evaluable on the public methylation backend at this revision; this section is a workflow audit (the pre-registered transport gate worked; the decisive control was correctly excluded rather than absorbed into the score) rather than a validation.** Full transport audit, secondary-backend scan, T9 hypothetical (~18-fold separation under Roth's BSS polarity, explicitly **not** counted as validation), retained directional-pair scores, and Endpoint-2 target-side editability check are in PAPER.md Appendix B. Reproduce with `uv run python scripts/score_roth_transport_subset.py`.

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6 Discussion

6.1 What V2.5 is and is not

On matched cell-line cohorts — where the PAM cytosine's methylation state cleanly separates the tumor and normal arms — raw $\Delta\beta$ is already a strong baseline and V2.5 improves on it by a consistent but small margin (+0.010 to +0.080 across all nine tier × path rows on GSE322563 and GSE77348). **V2.5 does not dominate $\Delta\beta$ by a large AUC margin on easy cohorts, and we do not claim it does** — the practical case for V2.5 over $\Delta\beta$ on such cohorts is structural, not a matter of AUC supremacy. What V2.5 *does* do, relative to the published baselines ($\Delta\beta$-only, $\Delta\beta$_z, V1, V2 `tumor_only`, limma-style moderated-`t`), is summarized below:

1. **V2.5-sigmoid is the recommended probabilistic prioritization axis, supported by three precise claims rather than a blanket dominance:** (a) AUC-equivalent to V2.5-diff on all three matched cell-line cohorts (within 0.002 across GSE322563 HM450, GSE322563 native EPIC v2, and GSE77348; PAPER.md §5.2.2 WG panel); (b) strictly better WG top-K usability — `tie_band@100 = 1` for V2.5-

sigmoid vs 421 / 1,127 / 1,493 for V2.5-diff under the WG denominator on the three matched cell-line cohorts; (c) higher transported-label all-positive AUC on the single tissue cohort tested — V2.5-sigmoid beats V2.5-diff by +0.05 to +0.08 on GSE69914 (0.862 vs 0.778 at the validated tier). The tissue gain is per-positive heterogeneous (PAPER.md §5.9: GATA3 carries the matched-pool signal, ESR1 is matched-near-random under V2.5-sigmoid; PAPER.md §5.7: in the EvidenceClass-restricted `EXACT + PROXIMAL_CLOSE` ESR1 universe, V2.5-sigmoid trails V2.5-diff / Δβ-only / limma-style). V2.5-sigmoid ships as `tumor_plus_gap_sigmoid` in this tag; V2.5-diff is retained for backward compatibility and AUC parity. (The OOD boundary case GSE68379 correctly inverts for all axes and is reported separately in §5.4; it is not pooled with the prioritization-mode comparisons here.) Row-by-row margins against the shipped baselines:

- V2.5 over Δβ-only on 9 matched-cell-line rows: **+0.010 to +0.080**, consistently positive.
- V2.5 over V1 on 9 matched-cell-line rows: **+0.014 to +0.169**, small on GSE77348 (~+0.015) and moderate-to-large on GSE322563 (up to +0.169 on HM450 validated). No single "typical" margin — the magnitude depends on the cohort.
- V2.5 over V1 on the 3 tissue rows: **+0.113, +0.172, +0.291** (validated, narrow, wide).
- No general order invariant across Δβ and V1 — both above and below V2.5 reshuffle relative to each other across rows (§5.2). The only invariants that hold 9/9 on matched cell-line are *V2.5 > Δβ* and *V2.5 > V1*.
- The raw-AUC winner on tissue is V2 `tumor_only` (0.803–0.874), but its `tie_band@100 = 6,540` puts its top-100 inside a massively tied region, disqualifying it as a prioritization axis (retained as a diagnostic for AUC cross-checking).

2. **V2.5 rescues tissue-cohort ranking that V1 and Δβ both lose.** V1 collapses on GSE69914 wide (0.435, at chance) and degrades sharply on narrow (0.539); Δβ-only collapses across tiers (0.435–0.591). The mechanism: GSE69914 is bulk breast tissue on both arms (sporadic tumors + healthy-donor tissue), and bulk-sample β values are cell-type mixtures rather than pure populations, which attenuates clean cell-line differentials to small noisy gaps at the same loci (§5.3). V2.5 recovers the ordering because its IQR-derived per-probe σ (meaningful at n = 305 tumor / n = 50 normal) lets `p_diff` weight each candidate's gap by its confidence — probes with large-but-noisy gaps get down-weighted versus probes with moderate-but-tight gaps. Raw Δβ has no such mechanism, and V1′s weighted-sum heuristic does not propagate per-probe uncertainty. The tie-band prediction from §2.4 also holds: V2.5′s band at K = 100 shrinks from 190 (`n = 2/2`) ⟶ 299 (`n = 3/3`) ⟶ 2 (`n = 305/50`) with no hyperparameter changes.
3. **V2.5 emits tie-band-honest top-K reporting.** `tie_band_size_at_k` and `precision_at_k_{min, max}` let a reader see when the top-K window sits inside a saturated region of the composite (the low-replicate case). Δβ and V1 have `tie_band = 1` by construction (continuous scores) so they do not need the explicit reporting; V2.5 needs it because of its discrete `EvidenceClass` saturation, and reporting the intervals is what makes its low-`n` top-K honest.
4. **V2.5 produces a probability-scale interpretation.** Each per-site *probability-scale selectivity score* ∈ [0, 1] is composable with downstream probabilistic inputs (target-mutation models, gRNA off-target probabilities, delivery-efficiency priors). Δβ is a signed similarity and V1 is a weighted-sum heuristic; neither is directly multiplied into such a chain.
5. **On the OOD boundary (GSE68379), both probabilistic axes and Δβ-only invert correctly** below 0.5 — the scorers are consistently identifying that the Roth label set does not transport to Sanger MCF-7. This is a sanity check, not a generalization claim.

This is the methodologically honest framing: Δβ is a strong baseline that should be reported alongside any new methylation-guided Cas9 scorer; `gap_sigmoid` (shipped in this tag as `tumor_plus_gap_sigmoid`) is the recommended probabilistic prioritization axis across every non-boundary cohort shape we tested,

supported by three precise claims rather than a blanket dominance: (1) AUC-equivalent to shipped V2.5 on all three matched cell-line cohorts (within 0.002, PAPER.md §5.2.2); (2) `tie_band@100 = 1` for `gap_sigmoid` vs 421–1,493 for shipped V2.5 across the three WG matched cell-line cohorts — V2.5′s top-K is unusable at WG scale on n = 2/2 or 3/3 cohorts while `gap_sigmoid`'s is clean; (3) higher transported-label all-positive AUC on the single tissue cohort tested, with explicit per-positive heterogeneity (GATA3 carries the matched-pool signal; ESR1 is matched-near-random under V2.5-sigmoid in PAPER.md §5.9, and within the EvidenceClass-restricted `EXACT + PROXIMAL_CLOSE` ESR1 universe V2.5-sigmoid trails V2.5-diff / Δβ-only / limma-style per PAPER.md §5.7). Shipped V2.5 is retained as a selectable mode for backward compatibility and AUC parity. V1 remains the stable-release default for reasons of backward compatibility and its deterministic `tie_band = 1` top-K — *not* because V1 wins AUC anywhere. §6.2 below states the shipping logic explicitly.

6.2 Decision hierarchy

Three truths coexist:

1. **V1 `final_score` is the stable-release default** (tagged `v0.4.0`). Its continuous deterministic score gives `tie_band = 1` on every cohort tested — top-K is never tie-break-dependent, regardless of replicate count or cohort shape. V1 is the default because of this property and for backward-compatibility with existing users, not because it wins AUC anywhere.
2. **The recommended probabilistic axis is `gap_sigmoid`** (`probabilistic_mode: tumor_plus_gap_sigmoid`, ships in this tag). PAPER.md §5.2.2 cross-cohort WG panel shows `gap_sigmoid` is AUC-equivalent to shipped V2.5 on matched cell lines (GSE322563 HM450 0.989 vs 0.989, native v2 0.998 vs 0.998, GSE77348 0.982 vs 0.982), has higher transported-label AUC on the single tissue cohort tested (GSE69914 0.862 vs 0.778), and critically has `tie_band@100 = 1` at WG scale on every matched cell-line cohort (vs V2.5′s 421–1,493 on the same WG catalogs, where V2.5′s top-K becomes unusable). The tissue advantage is per-positive heterogeneous (PAPER.md §5.9: GATA3 carries the matched-pool signal, ESR1 is matched-near-random under `gap_sigmoid`; PAPER.md §5.7: in the EvidenceClass-restricted `EXACT + PROXIMAL_CLOSE` ESR1 universe, `gap_sigmoid` trails V2.5-diff / Δβ-only / limma-style). Shipped V2.5 (`probabilistic_mode: tumor_plus_differential_protection`) is retained as a selectable mode for backward compatibility and AUC parity on cell-line cohorts but is no longer recommended over `gap_sigmoid` for new prioritization runs. For GSE68379 (OOD cross-series), no axis is supported.
3. **V2 `tumor_only` is retained as a diagnostic**, not a prioritization axis. Its tied bands at K = 100 are **5,271–14,914 across the five cohort paths tested** (GSE68379 5,271; GSE69914 6,540; GSE322563 HM450 10,005; GSE77348 11,848; GSE322563 native 14,914) — in every case large enough that ordering within the top-100 is determined by tie-break, not by score. Its raw AUC can lead (as on GSE69914 tissue, 0.803–0.874), but its top-K is not usable for target-shortlist construction.

The decision table below is the literal recommended hierarchy:

Use case	Recommendation
New probabilistic prioritization runs	**V2.5-sigmoid** (`tumor_plus_gap_sigmoid`). Hypothesis generation only; AUC-equivalent to V2.5-diff on matched cell lines (within 0.002), avoids V2.5-diff's low-`n` WG tied-band blowup (`tie_band@100 = 1` vs 421–1,493), higher transported-label AUC on the single tissue cohort tested (per-positive heterogeneous; see §5.4 / PAPER.md §5.7 / §5.9).
Backward-compatible probabilistic runs	**V2.5-diff** (`tumor_plus_differential_protection`). Retained for AUC parity with earlier scored JSONLs; not preferred for new WG-scale prioritization.
Stable deterministic top-K	**V1 `final_score`** (tagged `v0.4.0`). Continuous score with `tie_band = 1` by construction; retained for backward compatibility, not AUC leadership.
Diagnostic ablation	**V2 `tumor_only`**. Useful AUC sanity check, but top-K collapses into thousands-record tied bands (5,271–14,914 across the five cohort paths tested).
Cross-series label transport (GSE68379-like)	**Unsupported**. Cell-line drift can invert label logic; do not pool with prioritization-mode benchmarks.

6.3 Limitations

1. **Low-replicate tie bands.** On cohorts with `n < 30`, top-K is reported as an interval via `precision_at_k_{min, max}` and `tie_band_size_at_k`. The problem dissolves at `n ≥ 30` (demonstrated on GSE69914).
2. **Cell-line drift.** GSE68379 confirms that "MCF-7" is not a single epigenetic object across labs. Any V2.5 shortlist must be validated against the specific cell line intended for the follow-up editing assay.
3. **V2.5-diff σ_floor = 0.05** is empirical. A sweep over σ_floor ∈ {0.02, 0.05, 0.10, 0.15} (PAPER.md §5.3.1; `examples/sigma_floor_sweep.*`) shows the matched-cell-line AUC ranking is robust (≤0.02 movement across a 7.5×-range sweep) but tissue AUC peaks at σ_floor = 0.10, not the shipped default. This σ_floor sensitivity is a V2.5-diff property only; V2.5-sigmoid uses a fixed bandwidth `σ_fixed` (default ≈0.0707) and is not σ_floor-dependent. Formal regime-specific σ_floor default selection for V2.5-diff is deferred.
4. **Catalog scope.** Primary tables in §5.1 use chr5/6/10 candidates within ±500 bp of a methylation probe (the chromosomes carrying Roth Fig. 5d targets). The frozen whole-genome panel cited in the abstract (PAPER.md §5.2.2) extends the matched cell-line + tissue evidence to 19.8M HM450 (SHA256 `d20661c5…`) and 35.4M EPIC v2 (SHA256 `39df8f0f…`) candidates, and PAPER.md §5.9 adds within-chromosome feature-matched negative controls. Cross-chromosome and chromosome-class matched controls remain future work.
5. **SCREEN cCRE Registry** is not integrated; DNase-HS clusters are a proxy.
6. **System B transport failure.** The pre-registered HEK293T / HCT116 System B extension (§5.7; PAPER.md §5.10) does **not** upgrade the headline selectivity claim because *VEGFA* T9 — the editable-but-non-selective discriminator the pre-registration named — is low-coverage in the public ENCODE RRBS

backend and is not rescued by the secondary WGBS / EPIC v2 scan. The retained *EMX1* T4 / *PRDX4* T5 subset is a polarity diagnostic, not the planned T9-demotion test.

7. **No prospective wet-lab validation.** The claim is AUC + tie-band-aware top-K on public methylation data. All named top hits are unvalidated predictions.

6.4 Related work

Two bodies of prior art are adjacent but non-overlapping. Generic CRISPR guide-scoring tools (e.g. Azimuth, DeepCRISPR, CRISPRitz) score sequence properties and off-target risk; none of them incorporate methylation state at the PAM cytosine, because none of them target a methylation-sensitive Cas9 variant. Methylation-aware variant analysis tools (e.g. methylKit, minfi) provide per-probe differential analysis on arrays but do not produce ranked target lists; their output feeds a scorer, not replaces one. `thermocas` is the first open framework we are aware of that joins the two — a ranking scorer that consumes methylation array summaries and outputs tumor-selective ThermoCas9 target shortlists with honest uncertainty. Probe-level differential-methylation callers themselves are not excluded as comparators: a limma-style moderated-`t` baseline (Smyth 2004 empirical-Bayes variance prior, pure-Python implementation with R `limma::lmFit + eBayes` parity) is reported alongside V2.5 across every cohort × tier in §5.6, where it is competitive on matched cell lines but collapses to AUC 0.573 on the GSE69914 tissue cohort under transported Roth labels.

6.5 Next steps

1. Prospective wet-lab validation of 3–5 top V2.5-sigmoid candidates on MCF-7 vs MCF-10A (β-pyrosequencing + ThermoCas9 editing readout). This is the terminal step that changes the scientific claim.
2. **Regime preset selector** on top of the shipped `tumor_plus_gap_sigmoid` mode (PAPER.md §6.3): a one-shot `regime` YAML field that picks (gap factor, σ_fixed, δ) defaults per cohort regime. Per §6.1, all non-boundary presets recommend V2.5-sigmoid (the `matched_cell_line` and `primary_tissue` presets differ only in the `EvidenceClass` ramp-n implied by sample count, not in the gap factor). A `backward_compat` preset that opts into V2.5-diff is retained for users needing AUC parity with pre-ag scored JSONLs.
3. **Cross-chromosome denominator controls** — the WG panel (§5.2.2) and within-chromosome feature-matched audit (PAPER.md §5.9) are complete; chromosome-class and random-chromosome matched controls remain as loosening axes.
4. **Canonical R `limma` parity** — completed at `memo-2026-04-22-ba`. Per-probe Spearman/Pearson `t_mod` ≥ 0.9997 across the three R-parity cohorts (GSE69914 tissue / GSE77348 / GSE322563 HM450) versus R `limma::lmFit + eBayes` 3.66.0 on identical sample-level β + group inputs; full results in PAPER.md §5.8 with per-cohort artifacts under `examples/r_limma_parity_*.{tsv,md}`.
5. A second independent-lab matched MCF-7/MCF-10A EPIC v2 cohort if one becomes public, to replicate the V2.5 headline at `n ≥ 3` rather than `n = 2`.
6. Formal SCREEN cCRE Registry integration when a non-challenged download path becomes available.

---

7 Conclusion

`thermocas` offers a *compositional probability-scale scoring skeleton* (`p_targ × (gap factor) × p_trust`) paired with a *tie-band-aware benchmarking contract* for methylome-guided ThermoCas9 target prioritization. The compositional skeleton is the durable contribution: it decouples three independently-replaceable questions ("is the tumor side targetable?", "is the tumor-vs-normal gap large enough?", "is the observation trustworthy?"), and the §5.2.1 ablation + §5.2.2 cross-cohort whole-genome panel exercise that decomposition by swapping the gap factor between V2.5′s `p_diff` and a simpler fixed-bandwidth `gap_sigmoid`. The current shipped recommendation that emerges from those experiments is `gap_sigmoid` (`probabilistic_mode: tumor_plus_gap_sigmoid`, ships in this tag): AUC-equivalent to the earlier `tumor_plus_differential_protection` (V2.5) on matched cell lines, higher transported-label AUC on the

single tissue cohort tested, and substantially better low-n WG top-K usability. Shipped V2.5 is retained as a selectable mode for backward compatibility and AUC parity on cell-line cohorts but is no longer the recommended prioritization axis. V1 remains the stable-release default for its deterministic `tie_band = 1` top-K guarantee. Δβ-only is a strong matched-cell-line baseline that any methylome-guided Cas9 scorer should report alongside its own scoring axis, and we report it on every cohort × tier in §5.2. The combination is offered as an open research framework; groups extending Roth et al. to other cell-type pairs are the immediate target audience.

Data and code availability

- **Code**: https://github.com/AllisonH12/thermocas9, tagged `memo-2026-04-22-bv` for the exact revision evaluated here. BSD-3 licensed. `git rev-parse memo-2026-04-22-bv` resolves to a SHA in a fresh clone. Supersedes all prior same-day dated memo tags; see the PAPER.md tag ledger for per-tag provenance (what each tag added and what was corrected in the next one). A `scripts/verify_manuscript_claims.py` guard enforces a set of *selected known-risk* numerical claims — constants (`src/thermocas/probabilistic.py`), universal-quantifier wording, artifact counts, figure captions, and the tag-span claims most recently caught in reviewer cycles — against the committed bench JSONLs and package source. It is not a full table-AUC verifier; new numerical claims should be audited by hand or added to the `check_*` function set. Run it before cutting any subsequent tag.
- **Tests**: 245 passing under `uv run pytest -q`.
- **Cohorts**: public GEO series GSE322563, GSE77348, GSE69914, GSE68379. Build scripts in `scripts/build_gse*_cohort.py`.
- **Reference annotations**: UCSC hg19 `refGene.txt.gz`, `cpgIslandExt.txt.gz`, `rmsk.txt.gz`, and `wgEncodeRegDnaseClusteredV3.txt.gz` (fetched on demand from hgdownload.soe.ucsc.edu).
- **Positives**: `data/derived/positives_roth_{validated, narrow, wide}.txt` (HM450 path) and `data/derived/epic_v2_positives/positives_roth_*.txt` (native EPIC v2 path). Derived from Roth et al. Fig. 5d via Ensembl REST hg38 → hg19 liftover.
- **Benchmark artifacts**: every `BenchmarkResult` JSONL under `examples/*_roth_labels/` carries `precision_at_k`, `precision_at_k_{min, max}`, `recall_at_k`, `recall_at_k_{min, max}`, `roc_auc`, `tie_band_size_at_k`, and `tie_break_policy`.
- **Feature-matched controls**: `examples/feature_matched_negative_controls.{tsv,md}` reports the within-chromosome matched-negative audit used in PAPER.md §5.9.
- **Annotated top-20 TSV + Markdown shortlists** under `examples/*/top20_annotated_v25.{tsv,md}`.

Competing interests

The author declares no competing interests.

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9. MCF-7 / MCF-10A HM450 surrogate methylation series. **GEO accession** GSE77348.
10. Sporadic breast tumor + healthy-donor breast tissue HM450 methylation series. **GEO accession** GSE69914.
11. Sanger GDSC1000 breast cell-line panel HM450 methylation series. **GEO accession** GSE68379.

Appendix A · Reproducibility

The reproducer commands below assume the public raw inputs have already been staged under `data/raw/` (those files are gitignored because of their size; ~10 GB total). To stage them:

```
# GEO methylation series (one curl per cohort; ~5–8 GB total uncompressed)
mkdir -p data/raw/{gse322563,epic_v2,gse77348,gse69914,gse68379,ucsc}

curl -L -o data/raw/gse322563/GSE322563_beta_matrix_EPIC_v2.txt.gz \
  "https://ftp.ncbi.nlm.nih.gov/geo/series/GSE322nnn/GSE322563/suppl/GSE322563_beta_
matrix_EPIC_v2.txt.gz"
curl -L -o data/raw/gse69914/GSE69914_series_matrix.txt.gz \
  "https://ftp.ncbi.nlm.nih.gov/geo/series/GSE69nnn/GSE69914/matrix/GSE69914_series_
matrix.txt.gz"
# (GSE77348 and GSE68379 supplementary URLs follow the same NCBI GEO ftp pattern.)

# EPIC v2 platform annotation (Illumina GPL33022)
curl -L -o data/raw/epic_v2/GPL33022_family.soft.gz \
  "https://ftp.ncbi.nlm.nih.gov/geo/platforms/GPL33nnn/GPL33022/soft/GPL33022_family.
soft.gz"

# UCSC hg19 reference annotations
for f in refGene.txt.gz cpgIslandExt.txt.gz rmsk.txt.gz
wgEncodeRegDnaseClusteredV3.txt.gz; do
  curl -L -o "data/raw/ucsc/$f" "https://hgdownload.soe.ucsc.edu/goldenPath/hg19/
database/$f"
done
```

From a fresh clone of the repository at `memo-2026-04-22-bv` with the above raw inputs staged:

```
# One-time env setup
uv sync
uv run pytest -q           # 245 passing
```

```python
# Rebuild the EPIC v2 probe annotation on chr5/6/10 (~2 min, needs pyliftover)
uv run python scripts/build_epic_v2_probes.py \
  --soft data/raw/epic_v2/GPL33022_family.soft.gz \
  --output data/raw/probes_hg19_epic_v2.tsv

# Rebuild the GSE322563 native EPIC v2 cohort (~30 s)
uv run python scripts/build_gse322563_native_epic_v2_cohort.py \
  --beta-matrix data/raw/gse322563/GSE322563_beta_matrix_EPIC_v2.txt.gz \
  --epic-v2-probes data/raw/probes_hg19_epic_v2.tsv \
  --output-dir data/derived/gse322563_native_epic_v2_cohort/

# Score + benchmark one mode on one cohort (~90 s on a modern laptop)
uv run thermocas score-cohort \
  --cohort data/derived/gse322563_native_differential.yaml \
  --catalog data/derived/catalog_hg19_chr5_6_10.jsonl \
  --output data/derived/scored_gse322563_native_differential.jsonl

uv run thermocas benchmark \
  --scored data/derived/scored_gse322563_native_differential.jsonl \
  --positives data/derived/epic_v2_positives/positives_roth_validated.txt \
  --cohort-name GSE322563-native-validated-V2.5 \
  --score-field p_therapeutic_selectivity --top-k 20 \
  --no-enforce-holdout \
  --output examples/gse322563_native_roth_labels/bench_validated_differential.jsonl

# Annotate the top 20 with gene / CGI / repeat / DNase context + Markdown companion
uv run python scripts/annotate_top_hits.py \
  --scored data/derived/scored_gse322563_native_differential.jsonl \
  --top-k 20 \
  --rmsk data/raw/ucsc/rmsk.txt.gz \
  --dnase data/raw/ucsc/wgEncodeRegDnaseClusteredV3.txt.gz \
  --positives data/derived/epic_v2_positives/positives_roth_wide.txt \
  --output examples/gse322563_native_roth_labels/top20_annotated_v25.tsv \
  --markdown examples/gse322563_native_roth_labels/top20_annotated_v25.md
```

All tables and figures in this manuscript are reproducible from the committed artifacts alone — no re-scoring required.

Appendix B · Notes on this manuscript file

`MANUSCRIPT.md` is the Bioinformatics-submission-shaped version. The audit-trail memo that preserves the full development history (V2 misspecification narrative, pre-registration block, decision-table iterations, and the tag-stability discussion) remains as `PAPER.md` in the same repository. Both files point at the same underlying code and benchmark artifacts. Where a discussion point is elided in this manuscript for length, `PAPER.md` carries the longer form.